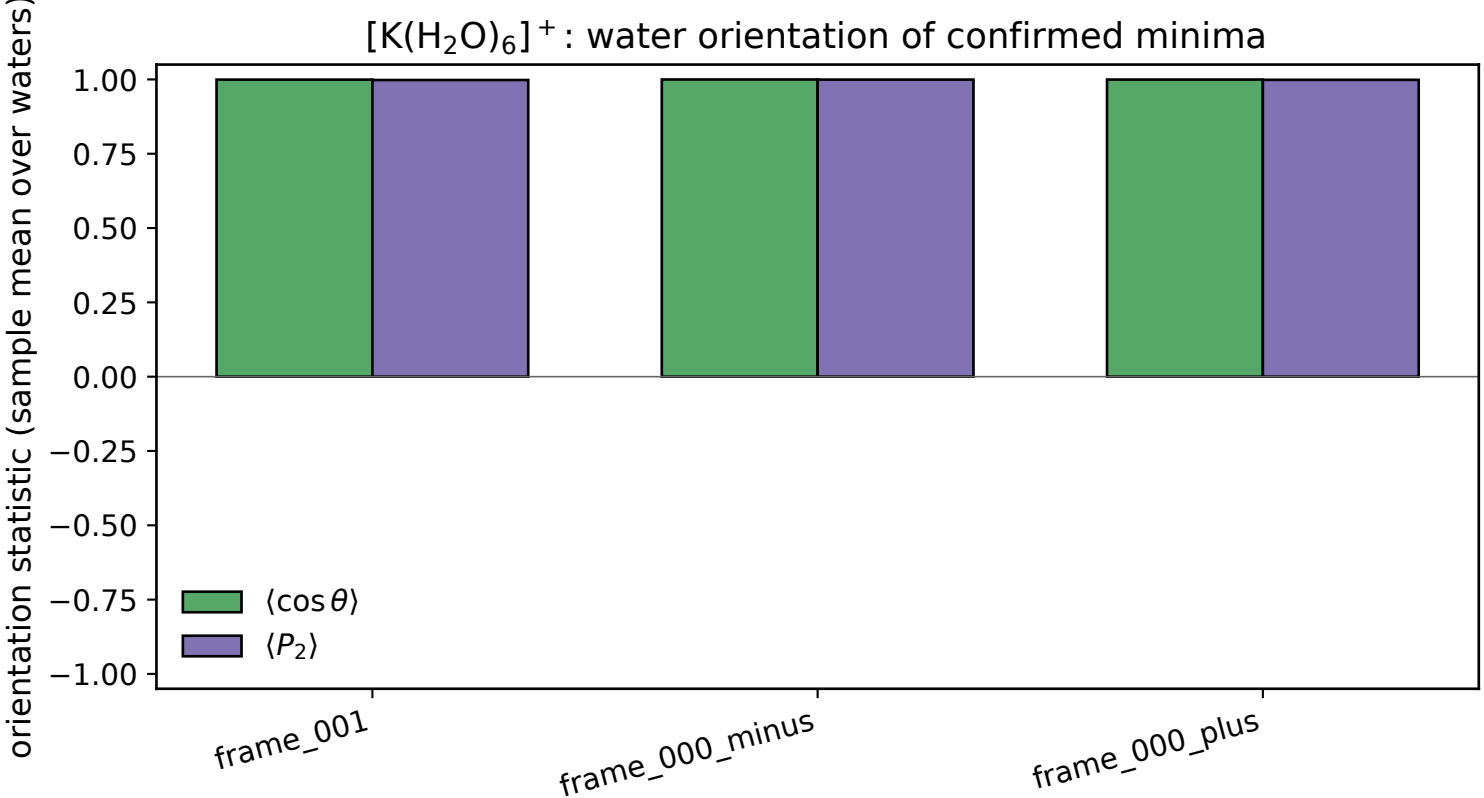

$\cos\theta$: angle(ion→O, O→H-midpoint). $P_2 = (3\cos^2\theta - 1)/2$. Unweighted over waters of one geometry.